

Model Order Reduction

Parametric PDE

Consider the FEM discretization of a PDE linear problem on Ω_h : find $u_h \in V_h$ such that

$$a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h, \quad +BC$$

and suppose that there are some parameters $\mu \in \mathbb{P} \subset \mathbb{R}^P$ that modify the problem:

- $\Omega_h(\mu)$
- $a(u_h, v_h; \mu)$
- $F(v_h; \mu)$
- $BC(\mu)$

We would like to

- know the solution for many μ_i (e.g. we want to compute statistics, we need to do optimization)
- we need a fast evaluation $\mu \mapsto u_h(\mu)$.

But it is **too expensive** to compute all FEM simulations

Model order reduction (MOR)

(or reduced order model (ROM))

We look for a smaller space $V_n = \langle \{\psi_i\}_{i=1}^n \rangle$ with dimension $n \ll N_h$, which is a **good approximation** of the space generated by $\langle \{u_h(\mu) \text{ with } \mu \in \mathbb{P}\} \rangle \subset V_h$, where $\mathbb{P}$ is the parameter space.

1. What does the problem becomes in the space V_n ?
2. Is solving the problem in V_n faster than in V_h ?
3. How do I find the space V_n ?

- FEM: $V_h = \langle \{\varphi_i\}_{i=1}^{N_h} \rangle$ and $u_h(x) := \sum_{j=1}^{N_h} \varphi_j(x) u_j$ with u_j the unknown coefficients.

$$\sum_j a(\varphi_i, \varphi_j; \mu) u_j = F(\varphi_i; \mu) \quad \forall i = 1, \dots, N_h$$

- ROM Galerkin Projection: $V_n = \langle \{\psi_i\}_{i=1}^n \rangle$ and $u_n(x) := \sum_{j=1}^n \psi_j(x) \hat{u}_j$ with u_j the unknown coefficients.

$$\sum_j a(\psi_i, \psi_j; \mu) \hat{u}_j = F(\psi_i; \mu) \quad \forall i = 1, \dots, n.$$

Matrix formulation

$\psi_i(x) = \sum_{j=1}^{N_h} B_{ji} \varphi_j(x)$. So, we can look at the ROM basis functions as a set of coefficients $B \in \mathbb{R}^{N_h \times n}$ that define the ROM space V_n as a subspace of V_h .

The FEM problem is given by $A(\mu)u = F(\mu)$, where $A_{ij}(\mu) = a(\varphi_i, \varphi_j; \mu)$ and $F_i(\mu) = F(\varphi_i; \mu)$.

The ROM problem is given by $\hat{A}(\mu)\hat{u} = \hat{F}(\mu)$, where

$$\hat{A}_{ij}(\mu) = a(\psi_i, \psi_j; \mu) = \sum_{k,l=1}^{N_h} a(B_{ki} \varphi_k, B_{lj} \varphi_l; \mu) = \sum_{k,l=1}^{N_h} B_{ki}(\mu) A_{kl}(\mu) B_{lj}(\mu) = (B^T A(\mu) B)_{ij}$$

and $\hat{F}_i(\mu) = F(\psi_i; \mu) = \sum_{j=1}^{N_h} B_{ji}(\mu) F_j(\mu) = (B^T F(\mu))_i$.

Full-Order Finite Element Model

$$Au = F$$

$$A \in \mathbb{R}^{N_h \times N_h} \quad u \in \mathbb{R}^{N_h \times 1} \quad F \in \mathbb{R}^{N_h \times 1}$$

N_h = dimension of the FE space (large)

Reduction Basis

$$B \in \mathbb{R}^{N_h \times n}$$

$N_h \gg n$
(tall and skinny)

Reduced-Order Model via Projection

Project full system:

$$B^T A B = B^T F$$

$$B^T A B \in \mathbb{R}^{n \times n} \quad \hat{u} \in \mathbb{R}^{n \times 1} \quad B^T F \in \mathbb{R}^{n \times 1}$$

n = dimension of the reduced space (small)

Galerkin
Projection

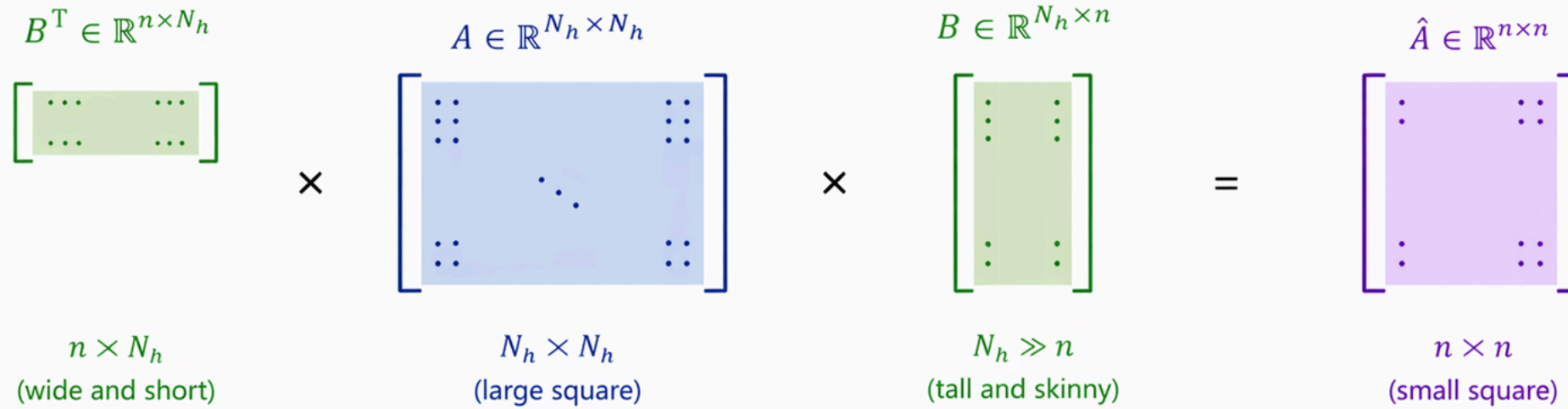
$$\hat{A} = B^T A B$$

$$\hat{F} = B^T F$$

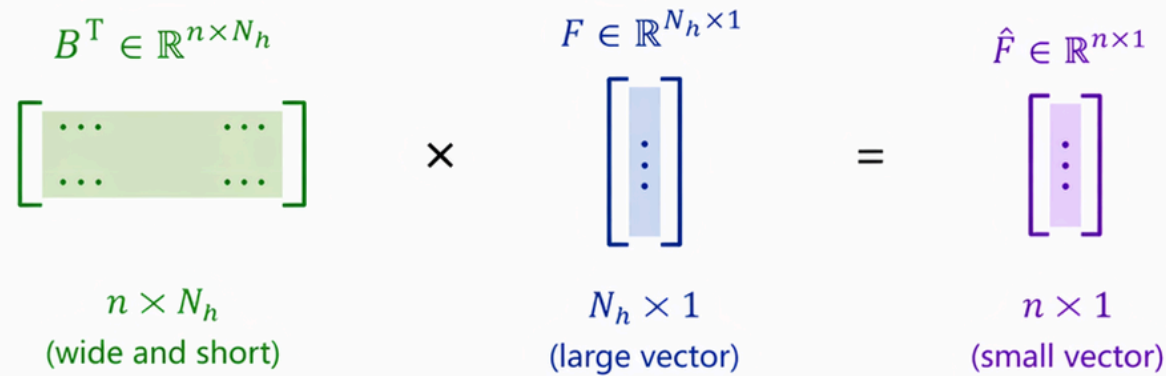
$$\hat{u} = B^T u$$

with $B^T B = I_n$ (orthonormal basis)

Matrix Projection: $\hat{A} = B^T A B$



Vector Projection: $\hat{F} = B^T F$



Affine decomposition of the system

The ROM problem is really cheaper if we can also assemble the ROM matrices ahead of time in a **offline** phase, i.e. if we can write

$$A(\mu) = \sum_{q=1}^{Q_a} \Theta_q^a(\mu) A_q, \quad F(\mu) = \sum_{q=1}^{Q_F} \theta_q^F(\mu) F_q,$$

where $A_q \in \mathbb{R}^{N_h \times N_h}$ and $F_q \in \mathbb{R}^{N_h}$ are parameter-independent matrices and vectors and $\Theta_q^a(\mu) : \mathbb{P} \rightarrow \mathbb{R}$ and $\theta_q^F(\mu) : \mathbb{P} \rightarrow \mathbb{R}$ are parameter-dependent scalar functions.

In this way, we can precompute the ROM matrices and vectors

$$\hat{A}_q = B^T A_q B, \quad \hat{F}_q = B^T F_q$$

and then assemble the ROM system in the **online** phase as

$$\begin{aligned} \hat{A}(\mu) &= \sum_{q=1}^{Q_a} \Theta_q^a(\mu) \hat{A}_q = \sum_{q=1}^{Q_a} \Theta_q^a(\mu) B^T A_q B = B^T A(\mu) B, \\ \hat{F}(\mu) &= \sum_{q=1}^{Q_F} \theta_q^F(\mu) \hat{F}_q = \sum_{q=1}^{Q_F} \theta_q^F(\mu) B^T F_q = B^T F(\mu). \end{aligned}$$

Example of affine decomposition: the Poisson problem with a parameter-dependent diffusion coefficient

Consider the Poisson problem with a parameter-dependent diffusion coefficient $\kappa(\mu)$:

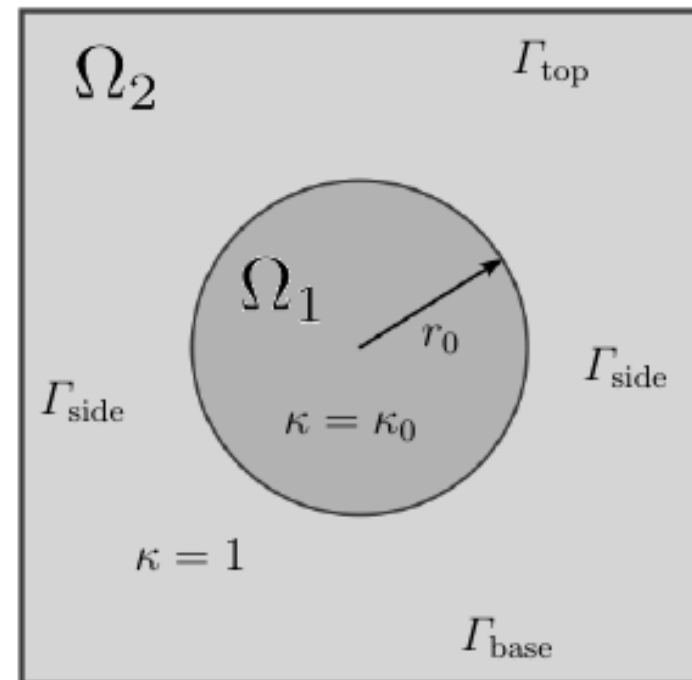
$$\begin{cases} -\nabla \cdot (\kappa(x; \mu) \nabla u) = f & \text{in } \Omega, \\ u(x) = 0, & x \in \Gamma_{top}, \\ \kappa(x; \mu) \nabla u(x) \cdot \mathbf{n} = 0, & x \in \Gamma_{side}, \\ \kappa(x; \mu) \nabla u(x) \cdot \mathbf{n} = \mu_2, & x \in \Gamma_{base}. \end{cases}$$

Where the diffusion coefficient $\kappa(\mu)$ is given by

$$\kappa(\mu) = \begin{cases} \mu_1 & \text{in } \Omega_1, \\ 1 & \text{in } \Omega_2. \end{cases}$$

The affine decomposition of a can be written as

$$\begin{aligned} a(\varphi_i, \varphi_j; \mu) &= \sum_{q=1}^2 \Theta_q^a(\mu) a_q(\varphi_i, \varphi_j), \\ a_1(\varphi_i, \varphi_j) &= \int_{\Omega_1} \nabla \varphi_i \cdot \nabla \varphi_j dx, & \Theta_1^a(\mu) &= \mu_1, \\ a_2(\varphi_i, \varphi_j) &= \int_{\Omega_2} \nabla \varphi_i \cdot \nabla \varphi_j dx, & \Theta_2^a(\mu) &= \mu_2, \\ f_1(\varphi_i) &= \int_{\Gamma_{base}} \varphi_i ds, & \theta_1^F(\mu) &= \mu_2. \end{aligned}$$



How to build the ROM space V_n ? Proper orthogonal decomposition (POD)

In an **offline** phase, we compute the solution of the full order model (FOM) for a set of parameters $\{\mu_1, \dots, \mu_{N_s}\}$ and we collect the solutions in a matrix $S = [u_h(\mu_1), \dots, u_h(\mu_{N_s})] \in \mathbb{R}^{N_h \times N_s}$, called **snapshot matrix** ($N_s \ll N_h$).

We then apply the **SVD** algorithm to the snapshot matrix S and we select the first n left singular vectors as the ROM basis functions $\{\psi_i\}_{i=1}^n$.

$$S = U\Sigma V^T, \quad U = [\psi_1, \dots, \psi_{N_s}] \in \mathbb{R}^{N_h \times N_s}, \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_{N_s}) \in \mathbb{R}^{N_s \times N_s}, \quad V = [v_1, \dots, v_{N_s}] \in \mathbb{R}^{N_s \times N_s},$$

with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{N_s} \geq 0$ the singular values of S .

Keep only the first n singular values, which gives the most important information of the snapshot matrix, and discard the rest.

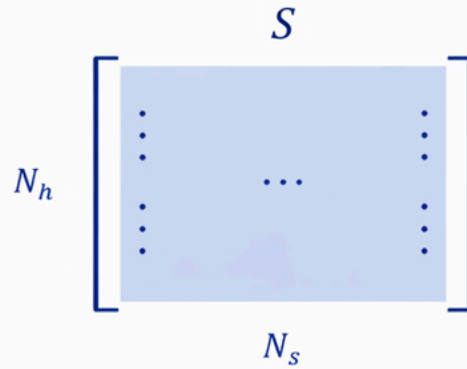
So, we approximate

$$S \approx \hat{S} = \hat{U}\hat{\Sigma}\hat{V}^T, \quad \hat{U} = [\psi_1, \dots, \psi_n] \in \mathbb{R}^{N_h \times n}, \quad \hat{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n) \in \mathbb{R}^{n \times n}, \quad \hat{V} = [v_1, \dots, v_n] \in \mathbb{R}^{N_s \times n}.$$

The ROM basis functions are given by $\psi_i = B_{:,i} := U_{:,i}$ for $i = 1, \dots, n$.

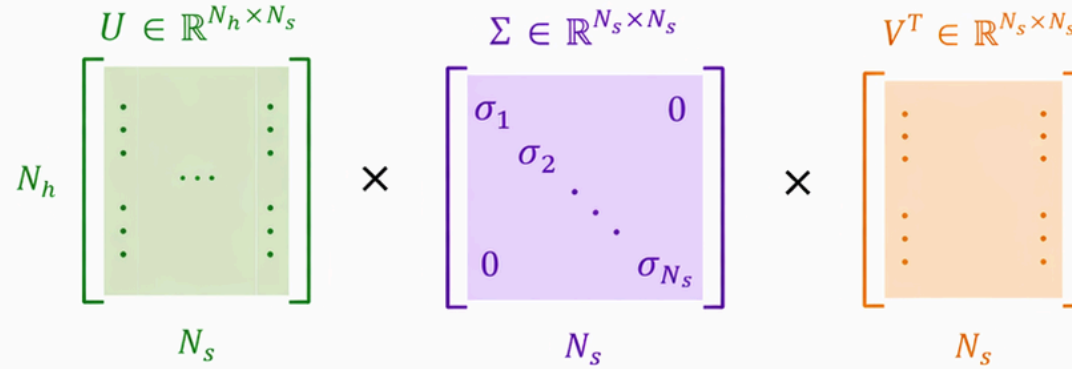
Given data matrix

$$S \in \mathbb{R}^{N_h \times N_s}$$



Singular Value Decomposition (SVD)

$$S = U \Sigma V^T$$



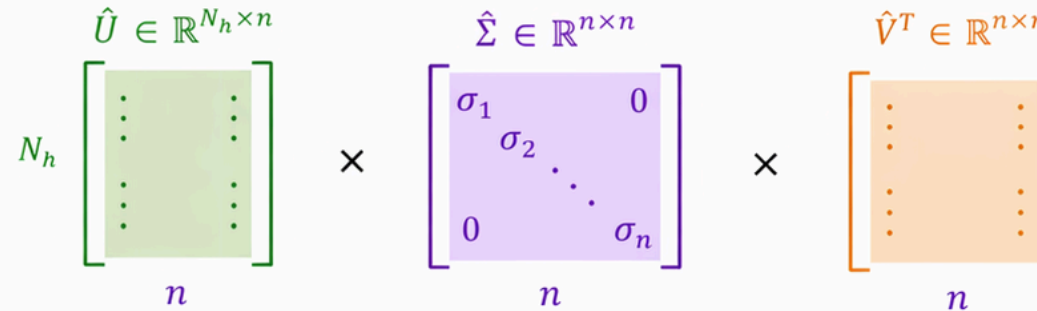
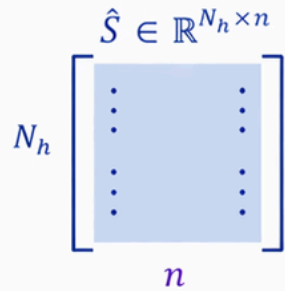
Dimensions:

- $U \in \mathbb{R}^{N_h \times N_s}$
- $\Sigma \in \mathbb{R}^{N_s \times N_s}$
- $V^T \in \mathbb{R}^{N_s \times N_s}$

(with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{N_s} \geq 0$)

Truncated SVD (rank- n approximation)

$$\hat{S} = \hat{U} \hat{\Sigma} \hat{V}^T$$



Truncate to retain only the first n singular values
($n \ll N_s$)

Dimensions after truncation:

- $\hat{U} \in \mathbb{R}^{N_h \times n}$
- $\hat{\Sigma} \in \mathbb{R}^{n \times n}$
- $\hat{V}^T \in \mathbb{R}^{n \times n}$

(keeps the first n singular values)

Best rank- n approximation: $\hat{S} = \operatorname{argmin}_{X \in \mathbb{R}^{N_h \times n}} \|S - X\|_F$ (Eckart-Young-Mirsky theorem)

Is it the best approximation? Theorem (Eckart-Young-Mirsky)

Given $S = U\Sigma V^T$ the SVD of S , the best rank- n approximation of S in the Spectral/Frobenius norm is given by

$$\hat{S} = \hat{U}\hat{\Sigma}\hat{V}^T, \quad \hat{U} = [\psi_1, \dots, \psi_n] \in \mathbb{R}^{N_h \times n}, \quad \hat{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n) \in \mathbb{R}^{n \times n}, \quad \hat{V} = [v_1, \dots, v_n] \in \mathbb{R}^{N_s \times n}.$$

with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{N_s} \geq 0$ the singular values of S .

This means that for any other matrix B of rank n we have

$$\|S - \hat{S}\|_2 \leq \|S - B\|_2, \quad \|S - \hat{S}\|_F \leq \|S - B\|_F.$$

Proof: https://en.wikipedia.org/wiki/Low-rank_approximation

Extension to time dependent problems

$$(\partial_t u_h, v_h) + a(u_h, v_h; \mu) = F(v_h; \mu) \quad \forall v_h \in V_h, \quad +BC$$

As before, we can for the reduced coefficients $\hat{u}(t; \mu)$ in the ROM space V_n :

$$\sum_j (\psi_j, \psi_i) \partial_t \hat{u}_j + \sum_j a(\psi_j, \psi_i; \mu) \hat{u}_j = F(\psi_i; \mu) \quad \forall i = 1, \dots, n, \quad +BC$$

Either for explicit or implicit time integration. Clearly, there is more advantage and reduction in the implicit case.

Linear acoustics problem

In the linear acoustics problem there are several questions open:

- Which parameters should we consider? (Time is one parameter, Initial conditions can be parametrized, the speed of sound can be parametrized)

$$\begin{cases} \partial_t p + \rho_0 c^2 \nabla \cdot \mathbf{u} = 0, \\ \partial_t \mathbf{u} + \frac{1}{\rho_0} \nabla p = 0. \end{cases}$$

- How do we consider the ROM? Component by component? Or all together? i.e., are we looking for a ROM space such that

$$(p(x, \mu), u(x, \mu), v(x, \mu)) = \sum_{i=1}^n (\psi_i^p(x), \psi_i^u(x), \psi_i^v(x)) \hat{c}_i$$

or

$$p = \sum_{i=1}^{n_p} \psi_i^p \hat{p}_i, \quad u = \sum_{i=1}^{n_u} \psi_i^u \hat{u}_i, \quad v = \sum_{i=1}^{n_v} \psi_i^v \hat{v}_i?$$

- Will the scheme be stable? In the explicit case? In the implicit? Can we say something on energy stability?